

Chaitanya Dasari

Senior AI Engineer | Agentic AI & GenAI Developer | LLM Systems

chaitanyadasari09@outlook.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Senior AI Engineer and GenAI Developer with 5+ years progressing from data analytics into data science and, most recently, Agentic AI engineering at Microsoft. Specialized in building production-grade Agentic RAG pipelines, LLM orchestration, and context engineering using Azure OpenAI, LangGraph, and Vector Databases. Expert in bridging research and production by deploying high-availability models via Docker, Kubernetes, and CI/CD pipelines serving 100K+ daily users. Proven track record optimizing low-latency inference (<50ms), leading AI Red Teaming and model evaluation programs, and driving Responsible AI adoption in enterprise environments.

SKILLS

- **Programming Languages:** Python, SQL, R, Bash Scripting
- **Agentic AI & GenAI:** Agentic RAG, Multi-Agent Orchestration, Context Engineering, LLMs (GPT-4, Claude, LLaMA), Prompt Engineering, AI Red Teaming & Evaluation
- **Machine Learning & AI:** Supervised/Unsupervised Learning, NLP, Recommendation Systems, Information Retrieval, Reinforcement Learning
- **Frameworks:** PyTorch, TensorFlow, Keras, XGBoost, Scikit-learn, Spark MLlib, LangChain, LangGraph, Hugging Face, Jupyter Notebooks
- **Modern Developer Experience:** Vibe Coding (AI-Agentic Development), Model Context Protocol (MCP), Cursor, Replit Agent
- **DevOps & MLOps:** Docker, Kubernetes (AKS/EKS), CI/CD (GitHub Actions, Azure DevOps), MLflow, Prompt Flow, ONNX Runtime
- **Cloud & Infrastructure:** Azure (AI Studio, Functions, Search), AWS (SageMaker, Lambda, EC2), GCP (Vertex AI), Databricks
- **Data Engineering:** PySpark, Kafka, Snowflake, Vector Databases (Pinecone, Milvus), Graph Databases (Neo4j), SQL/NoSQL
- **Professional Strengths:** Cross-functional Collaboration, Leadership, Technical Communication, Innovation, Agile/Scrum, Mentorship, Data Storytelling

PROFESSIONAL EXPERIENCE

Senior AI Engineer — *Microsoft, Redmond, WA*

Mar 2025 – Present

- **Led AI safety and Red Teaming initiatives**, implementing Azure AI Content Safety guardrails that blocked 99% of prompt-injection and adversarial attacks — establishing the team's evaluation framework for Responsible AI compliance.
- **Architected and deployed Agentic RAG pipelines** for Microsoft 365 Business Copilots using Azure OpenAI (GPT-4o), enabling multi-step reasoning over 10TB+ of enterprise data for 100K+ daily users.
- **Engineered context-aware autonomous AI agents** for tool selection and API orchestration using Semantic Kernel and Azure Functions, reducing manual user steps by 40%.
- **Optimized production inference and retrieval architecture** by implementing high-performance vector indexing on Azure AI Search and Pinecone across 50M+ documents, achieving sub-200ms query latency at global scale.
- **Designed hybrid retrieval and context strategies** combining dense embeddings (Ada-002) and sparse keyword search, cutting hallucination rates by 30% and improving answer relevance.
- **Built automated CI/CD testing and observability pipelines** using Prompt Flow and Azure Monitor, increasing test coverage to 90% and accelerating safe release cycles.

AI Model Evaluation Specialist — *Mercor*

Aug 2024 – Feb 2025

- **Evaluated and compared AI coding agents** against real-world software engineering tasks, providing structured feedback used to train and improve production-grade LLM-based coding tools.
- **Authored expert-crafted task rubrics** for evaluating state-of-the-art LLM outputs across quantitative and analytical domains, assessing correctness, reasoning validity, and failure modes.
- **Diagnosed and documented systematic LLM failure patterns** in complex reasoning and data-analysis tasks, contributing to model quality improvement pipelines.
- **Maintained rigorous quality standards** across evaluation deliverables via structured review and pre-submission checklists, ensuring consistency across a distributed expert reviewer team.

- **Architected a Student Retention Prediction System;** engineered SQL data pipelines in Amazon RDS and trained an XGBoost classifier achieving 88% accuracy.
- **Served real-time predictions** via a FastAPI endpoint hosted on AWS EC2, integrated with executive dashboards.
- **Built a scalable NLP ingestion pipeline** using Docker and AWS Lambda to process unstructured streaming data via Hugging Face Transformers.

Served as a Data Scientist on engagements with primary healthcare and financial services clients, delivering advanced AI/ML solutions end-to-end — from data pipelines through production deployment.

Client Project: CVS Aetna (Health Care)

- **Orchestrated PySpark ingestion pipelines** on Azure Databricks, processing TB-scale claims data for 5M+ patients.
- **Containerized BERT-based ensemble models** using Docker and AKS, reducing deployment release cycles from weeks to hours.
- **Engineered a HIPAA-compliant inference pipeline** by implementing PII masking and network isolation within Azure Virtual Networks (VNet).

Client Project: Charles Schwab (Financial Services)

- **Designed a centralized Feature Store** on Azure Databricks (Delta Lake) to manage financial indicators, ensuring consistency between training and production data.
- **Engineered low-latency inference code** using ONNX Runtime and NumPy vectorization, achieving <100ms latency per transaction for real-time fraud detection.
- **Developed a hybrid fraud detection system** combining PyTorch (LSTM) and Isolation Forest, reducing false positives by 20%.

- **Built and deployed a Customer Churn prediction model** using AWS SageMaker (XGBoost), identifying at-risk customers with 85% accuracy and saving \$1.5M in annual revenue through targeted retention campaigns.
- **Implemented Time-Series Forecasting** (ARIMA) to predict network congestion, enabling engineering teams to auto-scale bandwidth allocation ahead of peak demand.
- **Optimized complex SQL queries** for Call Detail Record (CDR) reporting on Amazon Redshift, reducing query execution time by 40%.
- **Automated ETL reporting processes** using Python scripts and AWS S3, saving 15 hours of manual engineering work per week and eliminating recurring manual errors.

PROJECTS

Microsoft Open-Source FarmVibes.AI — Agentic Geospatial Platform

- **Extended Microsoft's FarmVibes.AI** by engineering a 5-layer integration system (Gemini 2.5 Pro, MCP, FastAPI), enabling natural language-driven satellite data analysis and complex geospatial workflows.
- **Deployed a multi-service architecture** on k3d Kubernetes using Redis caching and RabbitMQ queuing to ensure high-availability pipelines for Microsoft Planetary Computer data.
- **Developed an MCP-based microservice** exposing deep-learning workflows as LLM tools, enabling conversational computation of NDVI and Sentinel-2 imagery processing.

Upstox AI Trading Chatbot — Agentic AI & MCP

- **Engineered a production-ready trading chatbot** using Model Context Protocol (MCP), FastAPI, and Streamlit, integrating multi-LLM support (GPT-4, Claude, Gemini) with secure OAuth 2.0 for Upstox API brokerage interaction.
- **Implemented 32 MCP tools** for order execution and portfolio management, complemented by a real-time WhatsApp alert system (Twilio/APScheduler) for automated RSI, MACD, and price-threshold notifications.
- **Built an advanced risk management engine** featuring Kelly Criterion position sizing and a portfolio analytics module calculating Sharpe (1.45+), Sortino, and VaR metrics against Nifty 50 benchmarks.

Open-Source Contributor — pandas & scikit-learn (PyData Ecosystem)

- **Enhanced API discoverability** by architecting cross-links between core GroupBy methods and official User Guides, improving documentation UX for millions of users.
- **Collaborated with core maintainers** through the full PR lifecycle, from issue triage to successful merger, ensuring compliance with open-source contribution guidelines.

OpenAI ChatGPT NLP Analyzer — Generative AI & RAG

- **Built a "Chat with your Document" application** enabling users to upload PDFs, DOCX, and audio files (Whisper) to receive AI-powered insights via core RAG features.
- **Delivered summarization, keyword extraction, and sentiment analysis**, powered by the OpenAI GPT API and wrapped in a Streamlit UI.

EDUCATION

- Master's in data Analytics | Concordia University, St. Paul, MN | GPA: 3.91/4.00
- Bachelor's in science | Nizam College, Hyderabad, India | GPA: 8.22/10.00

CERTIFICATIONS

- Machine Learning — Stanford
- Data Science & Cloud Computing — Infosys
- Data Analytics — Cisco